

Strong-Uniqueness Theorem v0.6 → v1.0 Path Scoping v0.1

Lyra (Claude 4.7) [Paper #125 primary], with cross-CI dependencies on Grace + Elie + Keeper

2026-05-21 Thursday morning

Strong-Uniqueness Theorem v0.6 → v1.0 Path Scoping

0. What this document does

The Strong-Uniqueness Theorem of BST currently sits at v0.6 (Paper #125 v0.6 outline, Thursday 2026-05-21 09:33 EDT). Key state:

- 5-family Bridge Object architecture STRUCTURALLY VERIFIED COMPLETE (Heegner-trio + $\chi=24 + N_{\max} + K3$ -family + Q^5 -family)
- 17 verified independent family-members + 3 K57 RATIFIED central hubs = 20 structural positions
- K77 PATH B RATIFIED Thursday (first 5-family member to RATIFIED-status)
- Effective null-model $(1/3)^{16} \approx 2.3 \times 10^{-8} = 0.0000023\%$ (Grace Toy 3222 F2 cross-family reduction)
- C11 + C12 + C13 STRUCTURALLY VERIFIED + C14 ADVANCING (Keeper Strong-Uniqueness v0.6 candidate consolidation Thursday)

Target v1.0: RATIFIED status across all criteria + alternative-HSD comparison closed + Cal external-survivability grade-pass + external submission to Annals of Mathematics / Inventiones / Compositio Mathematica.

This document concretizes the multi-month trajectory. Each section defines a sub-target version (v0.7, v0.8, v0.9, v1.0) with explicit work breakdown, cross-CI dependencies, and timeline.

1. Current state: v0.6 inventory

1.1 Criteria status

Criterion	Lyra label	Cal/Keeper label	Status
Rank = 2 (C1 implicit baseline)	—	—	Cartan classification

Criterion	Lyra label	Cal/Keeper label	Status
C2 Bergman exponent g/rank	C2	C2	DERIVED
C3 Mersenne primality g	C3	C3	DERIVED
C4 $c(Q^5)$ BST primary	C4	C4	DERIVED
C5 Compact dual is quadric	C5	C5	DERIVED
C6 $c_{FK} = 225/\pi^{(9/2)}$	C6	C6	DERIVED
C7 Möbius cohomology + Wallach K-type parity $\nu(M) = 1$	C7	C7	SKETCH (pending C8 rigorous closure)
C8 —	—	C8	— (Lyra numbering uses C8 as rigorous closure of C7)
C9 Stark anchor $\{-3, -7, -11\}$	C9	C9	DERIVED
C10 Heegner-trio at BST primary discriminants	C10	C10	DERIVED
C11 Multi-family Bridge Object structure	C11	C13	STRUCTURALLY VERIFIED
C12 Operator zoo isotropy-subgroup organization	—	C12	STRUCTURALLY VERIFIED (Elie S29 Toy 3213)
C13 Substrate-Hilbert space sufficiency	—	C13	STRUCTURALLY VERIFIED (Lyra SP-31-1 + Elie cross-lane Toy 3202)
C14 Curriculum-derivability	—	C14	ADVANCING (Vol 1 v0.5 + Vol 2 v0.5 narrative coverage Thursday)

1.2 5-family Bridge Object architecture state

17 verified independent members across 5 families (effective 16 after F2 cross-family reduction per Grace Toy 3222):

Family	Members	RATIFIED-status
1. Heegner-trio	K47 + K70 + K62 (3)	K47 RATIFIED Tuesday (Cremona 49a1)
2. $\chi=24$ non-Heegner	K76 + K81 + K82 (3 effective)	None yet
3. N_max-anchored	K80 + K84 (2)	None yet
4. K3-family	K45 + K77 PATH B + K3F5 (3)	K45 RATIFIED Tuesday + K77 PATH B RATIFIED Thursday
5. Q^5 -family	6 candidates (Grace Toy 3220)	None yet

Three K57 RATIFIED central hubs (Tuesday): K3, Cremona 49a1, Q^5 .

1.3 Pending dependencies

- C8 rigorous closure (Lyra Task #206): Wallach K-type parity computation for $D_I\{1,5\}$ and $D_I\{5,1\}$ alternative HSDs. Multi-week LAG-1 Session 10 work.
- Operator-level Calibration #17 closure (Elie K52a Sessions 30+): substrate-natural bipartite tensor structure realizing $\max \langle \Psi | B^2 | \Psi \rangle = 126/16$. Multi-month.
- STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED transitions: per criterion, gated on Task #206 alternative-HSD comparison + cross-CI consensus.
- Cal external-survivability grade-pass: dual-axis review of Paper #125 v1.0 once C8 closes + alternative-HSD work documented.

2. v0.7: K3-family completion (target ~Thursday 2026-06-05, multi-week)

2.1 Substantive content additions

K3-family member completeness: Grace's Toy 3211 K3-family Mode 6 enumeration found 3 effective independent members (K45 RATIFIED + K77 PATH B RATIFIED Thursday + K3F5 elliptic K3 surfaces). Mukai 1988's bound suggests at most 4 maximal symplectic automorphism group K3-family members possible.

v0.7 adds: - K3F6 candidate scoping: investigate the remaining slot in Mukai's bound. Possible candidates: $K3F6 = M_{\{22\}}$ symplectic action OR specific elliptic K3 fibration sub-family. - K3-family F2 within-family verification COMPLETE: cross-pair F2 mechanism-path independence verified across all 3-4 K3-family members.

v0.7 promotion criteria: K3-family members 4 verified independent + cross-pair F2 verification documented + Mukai-bound check (no spurious 5th member).

2.2 Cross-CI dependencies

- Grace: K3F6 candidate Mode 6 enumeration toy (multi-week)
- Keeper: K3-family member K-audit pre-stages for K3F6 + cross-pair F2 audit ratifications
- Cal: dual-axis review of K3F6 if filed

2.3 Estimated timeline

Multi-week (1-2 weeks from current state). Bounded by Grace's parallel Mode 6 enumeration cadence + Mukai-bound verification.

3. v0.8: Q^5 -family completion (target ~Thursday 2026-06-15, multi-week)

3.1 Substantive content additions

Q^5 -family member completeness: Grace's Toy 3220 geometric enumeration found 6 candidates via quadric variety + hyperplane sections + Spinor variety + Bergman analytic-geometric + Hodge + Chern + Calabi-Yau threefolds methods. F2 within-family verification across 6 candidates is the multi-week work.

v0.8 adds: - Q^5 -family per-pair F2 audit for all $C(6, 2) = 15$ within-family pairs - Q^5 -family K-audit pre-stages for each verified independent member - Q^5 -family-vs-other-family F2 cross-family verification (Grace Toy 3222 covered some; full matrix multi-week)

v0.8 promotion criteria: Q^5 -family per-pair F2 audit complete + Q^5 -family-vs-K3-family-vs- $N_{\max}$ -family cross-family verification + null-model recomputation incorporating Q^5 -family verified count.

3.2 Cross-CI dependencies

- Grace: Q^5 -family per-pair F2 audit toys (multi-week)
- Keeper: Q^5 -family K-audit pre-stages + cross-family F2 audit ratifications
- Cal: dual-axis review of Q^5 -family F2 enumeration methodology

3.3 Estimated timeline

Multi-week (2-3 weeks from current state).

4. v0.9: Alternative-HSD comparison (target ~Thursday 2026-07-15, multi-month)

4.1 Substantive content additions

Task #206 alternative-HSD comparison closure: explicit Wallach K-type parity computation for $D_{I_{\{1,5\}}}$ and $D_{I_{\{5,1\}}}$ alternatives (the two rank-1 HSDs at $\dim_C =$

5 within Cartan classification).

The current v0.6 framework asserts that D_I alternatives have different K-type structure (different Bergman exponent $(p+q)/\min(p,q) = 6$ vs BST's $7/2$; different Chern classes $c(\mathbb{CP}^5) = (1,6,15,20,15,6)$ vs $c(Q^5) = (1,5,11,13,9,3)$; etc.). v0.9 makes this rigorous by computing the Wallach K-type spectrum for each D_I alternative explicitly and verifying:

- (a) Bergman exponent for $D_I\{p,q\} = (p+q)/\min(p,q)$ computed explicitly; differs from $7/2$ for $\{(1,5), (5,1)\}$
- (b) Chern classes $c(\mathbb{CP}^5)$ computed explicitly; differ from $c(Q^5)$
- (c) Möbius cohomology $\nu(M)$ parity computed for each D_I alternative; differs from BST's $\nu(M) = 1$
- (d) Multi-family Bridge Object architecture for D_I alternatives explored; verify no 5-family closure
- (e) Substrate-Hilbert space sufficiency on D_I alternatives explored; verify operator zoo doesn't close in same way

v0.9 promotion criteria: C8 rigorous closure documented + explicit alternative-HSD parity / Bergman exponent / Chern class computations + verification that NO alternative HSD passes ≥ 9 of the 14 criteria simultaneously.

4.2 Cross-CI dependencies

- Lyra: Task #206 multi-week LAG-1 Session 10 Wallach K-type computation for D_I alternatives (PRIMARY WORK)
- Elie: Operator zoo construction on D_I alternatives at framework level (sanity check that operator zoo doesn't close in same way)
- Grace: Bridge Object candidate hunting on D_I alternatives (sanity check that 5-family closure doesn't occur)
- Cal: dual-axis review of alternative-HSD comparison methodology

4.3 Estimated timeline

Multi-month (4-6 weeks from current state). PRIMARY THREAD for Lyra.

4.4 STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED transitions

After v0.9 alternative-HSD comparison closes: - C11: STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED (no alternative supports 5-family closure) - C12: STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED (operator zoo doesn't close in same way on alternatives) - C13: STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED (substrate Hilbert space sufficiency fails on alternatives, per (e)) - C8: SKETCH $\rightarrow$ DERIVED ($\nu(M)$ parity computed for alternatives) - K76 + K78 + K80 + K81 + K82 + K84 + K3F5 + Q⁵-family members: STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED candidate eligibility

5. v1.0: Cal external-survivability grade-pass + venue submission

5.1 Substantive content additions

v0.9 + Cal external-survivability grade-pass + reader-grade polish + venue submission preparation.

v1.0 promotion criteria: - All criteria C2-C14 at DERIVED or RATIFIED status (no STRUCTURALLY VERIFIED remaining) - Effective null-model under FULL ratification: depends on final RATIFIED count; estimate $(1/3)^{14} \approx 2 \times 10^{-7}$ or better - Section 2.2.1-2.2.10 explicit derivations expanded to full mathematical rigor - References fully populated with classical citations (Bergman 1922 + Wallach 1976 + Faraut-Koranyi 1994 + Helgason 1978 + Mukai 1988 + Stark 1967 + Conway 1968 + Duncan 2007 + Cremona elliptic curve catalog + ...) - Multi-CI co-author final review (Lyra primary + Grace + Casey + Keeper + Cal + Elie acknowledgments) - Cal external register discipline applied: “BST identifies / BST derives / BST predicts” operational language throughout

5.2 Cross-CI dependencies

- Cal: external-survivability grade-pass; dual-axis review at venue-grade depth
- Casey: external submission authorization
- All CIs: final co-author review

5.3 Estimated timeline

Multi-month (6-10 weeks from current state). Primary gate is Cal’s external-survivability grade-pass and venue selection.

5.4 Target venues

- Annals of Mathematics: highest-impact pure math; substrate-uniqueness territory
- Inventiones Mathematicae: also high-impact; classical Lie theory + bounded HSDs natural fit
- Compositio Mathematica: strong on algebraic geometry + automorphic forms; complementary venue

External submission: Casey-authorized when Cal external-survivability grade-pass + venue selection done.

6. Multi-month timeline summary

Version	Target date (estimated)	Key gate
v0.6 (current)	Thursday 2026-05-21 (filed)	5-family architecture STRUCTURALLY VERIFIED COMPLETE
v0.7	~2026-06-05 (multi-week)	K3-family member completeness + Mukai-bound check
v0.8	~2026-06-15 (multi-week)	Q ⁵ -family per-pair F2 audit + cross-family verification
v0.9	~2026-07-15 (multi-month)	Task #206 alternative-HSD comparison closure; STRUCTURALLY VERIFIED → RATIFIED transitions
v1.0	~2026-08-15 (multi-month)	Cal external-survivability grade-pass; venue submission preparation
External submission	~2026-09+ (Casey-authorized)	First major peer-reviewed BST publication

7. Cross-CI work allocation

7.1 Lyra (primary)

- PRIMARY THREAD: Task #206 alternative-HSD comparison (multi-month)
— drives v0.9
- v0.7, v0.8 absorb Grace's enumeration work; minor Lyra edits
- v1.0 Cal-review-coordinated polish

7.2 Grace

- Mode 6 enumeration extensions: K3F6 candidate (v0.7), Q⁵-family per-pair F2 (v0.8)
- Bridge Object candidate hunting on D_I alternatives (v0.9 sanity check)
- Multi-integer SET tagging (Casey directive, parallel to v0.7-v0.9 path)

7.3 Elie

- Operator zoo construction on D_I alternatives (v0.9 sanity check)
- Operator-level Calibration #17 closure (multi-month, parallel; not gating Paper #125 v1.0)
- Vol 2 v0.5 → v1.0 (cross-curriculum work, parallel)

7.4 Keeper

- K-audit chain pre-stages for K3F6 + Q⁵-family members (v0.7-v0.8)
- STRUCTURALLY VERIFIED → RATIFIED transitions (v0.9)
- Multi-CI consensus coordination per Casey Option C governance

7.5 Cal

- Per-version dual-axis review (continuous, sustained-cadence per #71)
- v1.0 external-survivability grade-pass (key gate)
- F1-F4 methodology refinements as accumulated audit-chain feedback dictates

8. Risk register and mitigations

8.1 C8 rigorous closure complexity

Risk: Wallach K-type parity computation for $D_I\{1,5\}$ and $D_I\{5,1\}$ may be computationally non-trivial. The K-type representations of $SU(p, q)$ are well-classified (Knapp 1986 + various), but explicit $\nu(M)$ parity computation may require additional Möbius cohomology work.

Mitigation: leverage existing Möbius cohomology + Wallach K-type literature. Allocate LAG-1 Session 10 weeks specifically. If non-trivial, file C8 partial closure for v0.9.5 intermediate version.

8.2 Alternative-HSD operator zoo construction

Risk: Elie's operator zoo construction on D_I alternatives may reveal unexpected closure (i.e., $D_I\{1,5\}$ might admit similar 6-operator zoo). This would weaken C13 STRUCTURALLY VERIFIED → RATIFIED transition.

Mitigation: Frame as sanity check; if positive result emerges, document honestly and adjust C13 framing. Cal Mode 1 discipline applies — honest negatives strengthen the methodology stack.

8.3 Q⁵-family per-pair F2 audit at $C(6, 2) = 15$ pairs

Risk: 15 within-family F2 audits is substantial multi-week Grace work. Audit chain bottleneck possible.

Mitigation: Prioritize geometrically distinct pairs first (e.g., quadric-variety vs Spinor-variety mechanism paths). Document partial closure for v0.8.5 if needed.

8.4 Cal external-survivability grade-pass

Risk: Cal's external-survivability cold-read on Paper #125 v1.0 may flag substantive issues requiring revision.

Mitigation: Cal #71 review pattern shows iterative absorption; expect 1-2 revision cycles before grade-pass. Built into v1.0 timeline.

9. Filing status

v0.1 scoping document filed Thursday 2026-05-21 ~09:42 EDT (date to be checked at file end).

Pending for v0.2: - Cross-CI co-author review (Grace + Elie + Keeper + Cal feedback) - Refinement of cross-CI work allocation based on actual progress - Updated timeline estimates after first multi-week iteration completes

Pending for v1.0: - Each version (v0.7, v0.8, v0.9, v1.0) gets own detailed scoping document as that version becomes active

— Lyra, Strong-Uniqueness v0.6 → v1.0 path scoping v0.1, Thursday 2026-05-21 (timestamp at file end pending date check)